

Group A

Assignment No: 9

Title of the Assignment: Data Visualization II

1. Use the inbuilt dataset 'titanic' as used in the above problem. Plot a box plot for distribution of age with respect to each gender along with the information about whether they survived or not. (Column names: 'sex' and 'age')
2. Write observations on the inference from the above statistics.

Objective of the Assignment: Students should be able to perform the data Visualization operation using Python on any open-source dataset

Prerequisite:

1. Basic of Python Programming
2. Seaborn Library, Concept of Data Visualization.

Theory:

1. Box Plot (Age vs Gender with Survival Information)

Using the inbuilt **Titanic dataset** (commonly from libraries like **Seaborn**), plot the required box plot-

Example-

```
import seaborn as sns
import matplotlib.pyplot as plt
# Load dataset
titanic = sns.load_dataset('titanic')
# Create box plot
plt.figure(figsize=(8,6))
sns.boxplot(x='sex', y='age', hue='survived', data=titanic)
plt.title('Age Distribution by Gender and Survival')
plt.xlabel('Gender')
plt.ylabel('Age')
plt.show()
```

2. Observations (Inference from the Plot)-

From the box plot, you can derive the following insights:

1. **Females had higher survival rates**
→ The survival category (1) for females shows a wider distribution compared to males.
2. **Younger passengers had better chances of survival**
→ Lower median age is visible in the survived group.
3. **Male survival rate is lower across most age groups**
→ The non-survived (0) category dominates for males.
4. **Children (especially girls) were prioritized**
→ Very young ages appear more in the survived category.
5. **Age spread is wider for non-survivors**
→ Indicates that people of all age groups among males mostly did not survive.

Based on these observations-

- The evacuation policy during the **Titanic disaster** followed a “**women and children first**” approach.
- This led to:
 - Higher survival rates among females
 - Increased survival probability for younger passengers
- Social norms and limited lifeboat capacity influenced survival patterns.
- Males, especially adults, were less likely to be rescued due to priority rules.

Conclusion

- Survival on the Titanic was strongly influenced by gender and age, where females and younger passengers had a significantly higher probability of survival due to priority-based evacuation strategies.